

ROS2 for ATOSFleetManagement Cheat Sheet

1) Source environment

```
```bash
source /opt/ros/humble/setup.bash
source ~/atos_ws/install/setup.bash
```
```

2) Build ATOSFleetManagement packages

```
```bash
cd ~/atos_ws
colcon build --packages-select atos atos_gui --symlink-install
source install/setup.bash
```
```

3) Start ATOSFleetManagement (normal)

```
```bash
ros2 launch atos launch_atosfleetmanagement.py insecure:=True
```
```

4) Start ATOSFleetManagement with 3 simulators

```
```bash
ros2 launch atos launch_atosfleetmanagement.py insecure:=True
with_truck_simulator:=True
```
```

Default simulator setup in launch

```
- `L5S-TRUCK-SIM-1`: `start_index=0`, `target_speed_kmh=80`,
`ignore_warning_speed_commands=True`
- `L5S-TRUCK-SIM-2`: `start_index=250`, `target_speed_kmh=40`
- `L5S-TRUCK-SIM-3`: `start_index=500`, `target_speed_kmh=40`
```

5) Open GUI

```
- URL: `http://localhost:8420`
- Go to tab: `RuralRoad Map`
- UI shows live trucks + `Distance To Next Truck Ahead`
```

6) Run in Docker (same image, with/without simulators)

```
```bash
cd ~/Documents/repos/ATOS
docker compose -f docker-compose-fleetmanagement.yml up --build
```
```

With simulators:

```
```bash
WITH_TRUCK_SIMULATOR=True docker compose -f docker-compose-fleetmanagement.yml up
--build
```
```

Without simulators:

```
```bash
WITH_TRUCK_SIMULATOR=False docker compose -f docker-compose-fleetmanagement.yml up
--build
```
```

Start without rebuild (if image already built):

```
```bash
docker compose -f docker-compose-fleetmanagement.yml up
```
```

Start in background:

```
```bash
docker compose -f docker-compose-fleetmanagement.yml up -d
```
```

...

When to use `--build`:

- Use `--build` only when image content changed (Dockerfile, dependencies, or source copied into image).
- For normal restart, do ****not**** use `--build`.

7) Run Docker as a systemd service

```
```bash
sudo mkdir -p /opt/atos
sudo rsync -a ~/Documents/repos/ATOS/ /opt/atos/
sudo cp /opt/atos/scripts/atosfleetmanagement.env.example
/etc/default/atosfleetmanagement
sudo cp /opt/atos/scripts/atosfleetmanagement.service
/etc/systemd/system/atosfleetmanagement.service
sudo systemctl daemon-reload
sudo systemctl enable --now atosfleetmanagement
```
```

Service controls:

```
```bash
sudo systemctl status atosfleetmanagement
sudo systemctl restart atosfleetmanagement
sudo journalctl -u atosfleetmanagement -f
```
```

8) Copy to another server (what files are needed)

Recommended (safest):

```
```bash
rsync -a ~/Documents/repos/ATOS/ user@<server>:/opt/atos/
```
```

Minimum required for Docker-based ATOSFleetManagement:

- `Dockerfile`
- `docker-compose-fleetmanagement.yml`
- `scripts/run\_atosfleetmanagement.sh`
- `scripts/installation/`
- `atos/`
- `atos\_gui/`
- `atos\_interfaces/`
- `conf/`
- Optional for systemd service:
 - `scripts/atosfleetmanagement.service`
 - `scripts/atosfleetmanagement.env.example`

9) Check running nodes

```
```bash
ros2 node list
```
```

10) Check key topics

```
```bash
ros2 topic list | rg "truck_objects|speed_command"
```
```

11) Watch live truck states for GUI

```
```bash
ros2 topic echo /atos/truck_objects/state
```
```

12) Watch control speed commands

```
```bash
```

```
ros2 topic echo /atos/truck_objects/speed_command
```
```

13) TruckObjectControl CoT interfaces

- ROS topic input: `/atos/truck\_objects/cot` (placeholder format)
- TCP input: `0.0.0.0:8114` listener in TruckObjectControl
- Truck clients connect to: `127.0.0.1:8114` (same machine) or `:8114`
- CoT `` is in `m/s` (GUI displays `km/h`)

14) Placeholder ROS CoT test (manual)

```
```bash
ros2 topic pub /atos/truck_objects/cot std_msgs/msg/String "data:
'id=truck_1;distance_m=1000;tcp_connected=1;lat=57.78357;lon=12.76389;speed_mps=2.8
;course_deg=150'" -1
```
```

15) Start one simulator manually (custom test)

```
```bash
ros2 run atos atos_truck_simulator --ros-args \
 -p uid:=L5S-TRUCK-UBUNTU \
 -p start_index:=300 \
 -p target_speed_kmh:=35.0 \
 -p acceleration_mps2:=0.6 \
 -p tcp_host:=127.0.0.1 \
 -p tcp_port:=8114
```
```

16) Simulator option: ignore first-limit slowdown

Use this only on selected trucks:

```
```bash
-p ignore_warning_speed_commands:=true
```
```

Behavior:

- Ignores non-zero warning command (for example `30 km/h` at `< 400 m`)
- Still obeys stop command (`0 km/h` at `< 200 m`)

17) Fleet control algorithm (stateful, per truck)

Initial rules (distance `X` to truck ahead on same path):

- If `500 m >= X > 200 m` => send `command=SLOWDOWN` (`target\_speed\_mps=8.333333`)
 - If `200 m >= X` => send `command=STOP` (`target\_speed\_mps=0.000000`)
 - If `X > 300 m` AND previous command was `STOP` => send `command=SLOWDOWN`
 - If `X > 700 m` AND previous command was `SLOWDOWN` => send `command=RESUME`
- (`target\_speed\_mps=nochange`)

Notes:

- Commands are evaluated per truck, based on its own truck-ahead distance.
- For each path, trucks are sorted by distance along trajectory before truck-ahead checks.

18) TCP speed command manual (per truck)

TruckObjectControl sends one newline-terminated command string per truck over the same TCP connection used by incoming CoT.

Fields:

- `command`: `STOP`, `SLOWDOWN`, or `RESUME`
- `target\_speed\_mps`: target speed in `m/s`, or `nochange` to keep current truck speed
- `distance\_to\_truck\_ahead\_m`: distance in meters from this truck to the next truck ahead in sorted trajectory order

- ``truck_ahead_path_index``: current path index for the truck ahead
- ``truck_ahead_uid``: UID for the truck ahead
- ``reason``: current command-state reason (``truck_ahead_stop_state``, ``truck_ahead_slowdown_state``, ``truck_ahead_resume_state``)
- ``min_gap_m``: smallest gap found among all connected/fresh trucks in current evaluation cycle
- ``connected_count``: number of trucks currently included in control logic
- ``path_name``: path used for the truck-ahead computation

Typical command strings:

```text

```
command=RESUME;target_speed_mps=nochange;distance_to_truck_ahead_m=812.432100;truck_
_ah
ead_path_index=310;truck_ahead_uid=L5S-TRUCK-SIM-
2;reason=truck_ahead_resume_state;min_gap_m=612.432100;connected_count=3;path_name=
RuralRoad_center_of_driving_lane_ccw.geojson
```
```

```text

```
command=SLOWDOWN;target_speed_mps=8.333333;distance_to_truck_ahead_m=312.500000;tru
ck_ah
ead_path_index=412;truck_ahead_uid=L5S-TRUCK-SIM-
3;reason=truck_ahead_slowdown_state;min_gap_m=154.270000;connected_count=3;path_nam
e=RuralRoad_center_of_driving_lane_ccw.geojson
```
```

```text

```
command=STOP;target_speed_mps=0.000000;distance_to_truck_ahead_m=92.140000;truck_ah
ead_path_index=418;truck_ahead_uid=L5S-TRUCK-SIM-
1;reason=truck_ahead_stop_state;min_gap_m=92.140000;connected_count=3;path_name=Rur
alRoad_center_of_driving_lane_ccw.geojson
```
```

19) Stop everything

- Press ``Ctrl+C`` in launch terminal